On the Magnetic Age of some Devonian Lavas from Devonshire

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ABSTRACT

The directions of magnetization of samples from two sites in the Devonian Ashprington Volcanic Series, Devonshire, are similar in spite of the difference in the dip of the lavas. They are close to the mean direction for the Keuper Marls and it is concluded that they resulted from remagnetization of the lavas during the Triassic.

DURING a palaeomagnetic survey of the basic igneous rocks of south-west England six samples were collected from two exposures of the Ashprington Volcanic Series, Devonshire. These are a series of lavas and, to a lesser extent, tuffs of Middle and Upper Devonian age exposed on both banks of the River Dart where they appear to be badly weathered and decomposed. The lavas are mainly fine grained spilitic basalts but normal augite basalts also occur. The presence of magnetite and ilmenite in the lavas has been noted by Ussher and Lloyd (1933) and unidentified threads of black dust are described as indicating fluxion structures in the schistose lavas.

Orientated samples of porphyritic spilitic lava were collected from riverside outcrops at Stoke Gabriel (N.G. 847568) and Greenway Ferry (N.G. 870549). The lavas at the former site are deep red in colour, due to an obviously secondary staining which has affected even the feldspar phenocrysts. This staining was not evident in the rocks at the second site.

The directions and intensities of the remanent magnetization were measured on an astatic magnetometer and the mean directions in Table 1 were found. The scatter of the sample directions about the mean has been calculated according to the statistics of Fisher (1953), in which κ is an estimate of the precision and $\propto$ is the half angle of the cone of confidence which includes the true mean with a probability of 0·05. The mean intensities and susceptibilities are given for each site. N is the number of samples measured.

The directions of the samples from the two sites are similar when corrected to the present horizontal, despite the fact that the difference in the dip of the lavas is 90 degrees, and are not comparable with the directions of other Devonian rocks. This suggests that the rocks have been remagnetized after folding for if they had retained their original directions of magnetization the closest agreement between different parts of the structure would occur when the beds were restored to their original horizontal. The directions at the two sites differ so markedly from the present geomagnetic field direction that a stable magnetization is suggested. There is one exception, a sample from Stoke Gabriel,

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which has a direction similar to that of the present field (Text-fig. 1). This has been omitted in computing the mean given in Table 1.

TABLE 1.—MAGNETIC DATA FOR DEVONIAN LAVAS FROM THE ASHPRINGTON SERIES, DEVONSHIRE

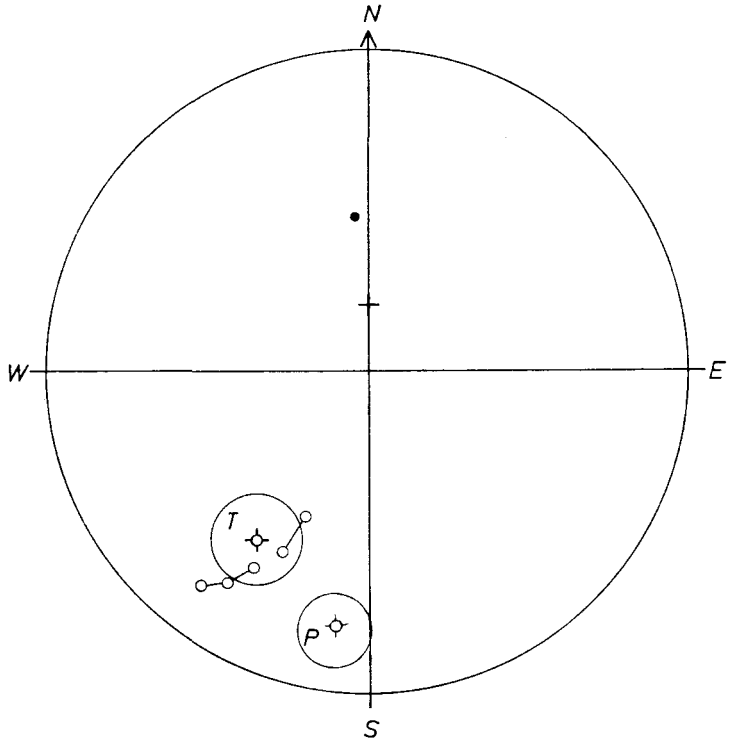
Site	*Directions of magnetization*		*Scatter*		*Intensity* $\times 10^{-6}$	*Susceptibility* $\times 10^{-6}$	*No. of samples*
	D°	I°	α°	κ	gauss	gauss/oersted	N
Stoke Gabriel	204	− 33	—	85	89	14	2(3)
Greenway	214	− 15	9	183	29	89	3
Mean	209	− 24	—	17	—	—	2

The age of the remagnetization can be estimated by comparing the observed directions of magnetization with other palaeomagnetic field directions for periods younger than the Devonian. In Text-fig. 1 the mean directions of the Exeter lavas (Upper Carboniferous or Lower Permian) (Cornwell, 1966) and the Keuper Marls (Clegg *et al.*, 1954) are shown together with their circles of confidence at the 95 per cent level. It is clear that there is a good agreement between the directions of the Devonian lavas and the mean Triassic result. The age of remagnetization can be even more precisely defined, for the Triassic was a period when the earth's magnetic field changed polarity several times (Clegg *et al.*, 1954). The Ashprington results indicate that the remagnetization occurred during a time when a reverse field existed.

The red staining observed in the Stoke Gabriel rocks must in some way be connected with the origin of the remagnetization. During thermal demagnetization experiments carried out to test the stability of the magnetization it was found that the remanence was not completely destroyed until temperatures of 600° C. had been exceeded. Curie temperatures of 660° C. were also measured for specimens from both sites with a thermomagnetic balance. These results indicate that a titaniferous haematite is present (the Curie temperature of pure haematite is 675° C.) and that it is responsible for at least part of the remanent magnetization. Two other Curie temperatures of 530° C. and 580° C. are considered to be due to magnetite, the presence of which is also indicated in the petrographic descriptions (Ussher and Lloyd, 1933). The directions of magnetization were not changed by thermal demagnetization or by a.c. demagnetization in fields of up to 800 oersteds. The Stoke Gabriel sample which has a direction similar to that of the present field (Text-fig. 1) was unfortunately also found to be resistant to demagnetization. It is therefore concluded that the present directions of the rocks at these two sites are due to remagnetization during the Triassic resulting from chemical changes in which haematite was formed. The red staining of the Stoke Gabriel samples is a visible indication of the presence of secondary haematite.

The Permo-Triassic is known from the work of several authors to

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TEXT-FIG. 1.—Directions of magnetization of samples from the Ashprington lavas, Devonshire, compared with the mean directions and circles of confidence of the Exeter lavas (P) and the Keuper Marls (T). Open and filled dots represent directions on the upper and lower hemispheres respectively. The present dipole field direction is shown by a cross.

have been a time when extensive weathering and reddening of pre-existing rocks occurred. Bailey (1926) concluded that the reddening of the Carboniferous rocks on Arran was developed *in situ* and was not the result of iron-bearing solutions percolating downwards. Trotter (1953) distinguished between red staining, which is due to the percolation of iron-bearing solutions and subsequent deposition of ferric oxide, partial oxidation and complete oxidation. The zone of complete oxidation arose on and below the ancient land-surface before the onset of the Permo-Triassic sedimentation and extends downwards to over 1,000 feet in places. Trotter pointed out that there is no difficulty in accounting for the presence of red-beds throughout the Carboniferous in north-west England but the difficulty lies in accounting for the vast quantities of iron remaining in a ferrous state. In north-east England

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there is a thickness of 25 to 50 feet of reddened Coal Measure strata lying immediately below the non-reddened basal sediments of the Permian. Anderson and Dunham (1953) suggested that the alteration is due to the *in situ* oxidation of pyrite and chalybite and to the introduction of iron oxides along joints and pore spaces.

Palaeomagnetic evidence for the remagnetization of older rocks due to oxidation during the Permo-Triassic periods has been described by Everitt and Belshé (1960). These authors found that many of the Carboniferous sediments from the Pennine area exhibited anomalous directions of magnetization which are best explained as the result of oxidation and remagnetization during the Triassic.

Cornwell (1966) has suggested that haematite, formed by the oxidation of the original magnetite in the Exeter lavas, Devonshire, is partly responsible for their remanent magnetization, which must then be regarded as a chemical or crystallization magnetization (CRM). The oxidation probably occurred soon after the lavas had been extruded in the Permian desert of south-west England. Chamalun and Creer (1964) and Creer (1966) also discovered Lower Palaeozoic rocks in south Wales and south-west England which had Permian directions of magnetization although the remagnetization in these cases was not connected with the Upper Palaeozoic climate. Creer (1964) has argued that widespread remagnetization of Lower Palaeozoic rocks occurred during the Permo-Carboniferous in the Laurasian continents of North America, Europe and Asia.

These results from Devonshire have been described because of the strong evidence, in the form of the directions of magnetization and the association of the remanence with haematite, suggesting oxidation during the Permo-Triassic. They also demonstrate that palaeomagnetic studies offer a suitable means of identifying the age of the oxidation.

This work was carried out in the Department of Geophysics and Planetary Physics, School of Physics, University of Newcastle upon Tyne.

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